

Problem 1:

Determine if the following statements describe a conceptual/logical, physical, or external representation/schema of data.

(a)

Employers can access an employee's name, job title, salary, and work schedule, but not their personal home address and bank account information.

■ c. external

(b)

All data is stored in tables sorted by category represented as bytes within a server hosted by AWS.

■ b. physical

(c)

The table "Students" has ID, Name, Date of Birth, Height, and Address fields. ID is unique for each item.

■ a. conceptual/logical

Problem 2:**(a)**

For the given instances (i.e. snapshot/subset) of a student relation, identify the following attributes (set) as might be a key or definitely not a key for the **entire** student relation, and provide explanations.

1. gpa might be a key since all gpa values are distinct in the table given.
2. sid might be a key since all id's are distinct in the table given.
3. first name is definitely not a key since there are two smith's.
4. {first name, contact} might be a key since the combination of these two attributes is unique across all rows in the table.

5.

List all single attributes that are definitely not candidate keys. Why are these attributes definitely not candidate keys? If all attributes could be candidate keys, explain why.

first name and age are definitely not candidate keys because there are duplicates in the snapshot. A candidate key must uniquely identify every tuple.

6.

Which single attribute is definitely a candidate key? Why? If there are no candidate keys, explain why.

While gpa and contact are also unique in this snapshot, sid is the only single attribute that is definitely a candidate key for the entire student relation. An ID is designed to be a unique identifier.

Problem 3:**(a)**

A nurse, Alice (eid=30), in a given shift, 12pm-8pm (sid=3), can care for more than one patient.

☐ True

(b)

The Nurse entity does not have a primary key.

☐ False

(c)

A patient, Dave (pid=23) cannot be “cared for” during multiple shifts by nurse Alice (eid=30).

☐ True

(d)

This design enforces that every patient is taken care of in each shift.

☐ IDK come back True

(e)

This design enforces that in each shift, at least one nurse is taking care of some patients.

☐ True

(f)

A patient, Dave (pid=23), can be taken care of by two different nurses in two different shifts.

☐ False

(g)

This design allows a patient that is not taken care of by any nurse.

☐ False

Problem 4:**(a)**

Does this design allow multiple musicians to collaborate on an album?

☐ No

(b)

Does this design allow multiple musicians to give concerts in the same place and date?

☐ Yes

(c)

Does this design allow musicians to collaborate with other musicians to perform a song? Musicians collaborate when they perform the same song, in the same place, on the same date.

☐ Yes

(d)

Does this design allow a musician to give multiple concerts in the same place on different dates?

☐ Yes

(e)

Does this design allow a musician to perform songs produced by another musician, in a concert?

☐ Yes

(f)

Does this design allow a musician to give a concert without performing any songs?

☐ No

(g)

Does this design allow for a song that is not in any album?

☐ Yes

(h)

Does this design allow a musician to produce an album that contains no songs?

☐ Yes

(i)

Does this design allow a musician to produce many albums?

☐ Yes

(j)

Does this design allow for a musician who only sings (i.e., does not play an instrument)?

☐ No

Problem 5:**(a)**

The fewest number of tables that can be used to represent this ER diagram without redundancy is 2

☐ True

(b)

In the least redundant design, 'mid' is a foreign key in the Employee table

☐ True

(c)

In the least redundant design, 'mid' is a foreign key in the Employee table

☐ True

(d)

In the least redundant design, 'mid' is a foreign key in the Employee table

☐ False

(e)

In the least redundant design, 'mid' is a foreign key in the Employee table

☐ False

(f)

Create a table for **Employee** by SQL statement, in the least redundant design, including all the necessary integrity constraints shown by the above ER diagram

```
CREATE TABLE Employee (  
    eid INTEGER PRIMARY KEY,  
    name CHAR(20),  
    age INTEGER,  
    mid INTEGER,  
    FOREIGN KEY (mid) REFERENCES Manager  
);
```